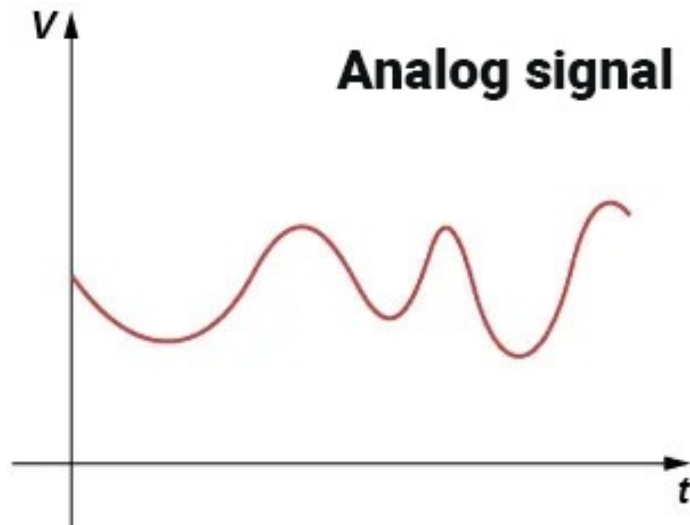


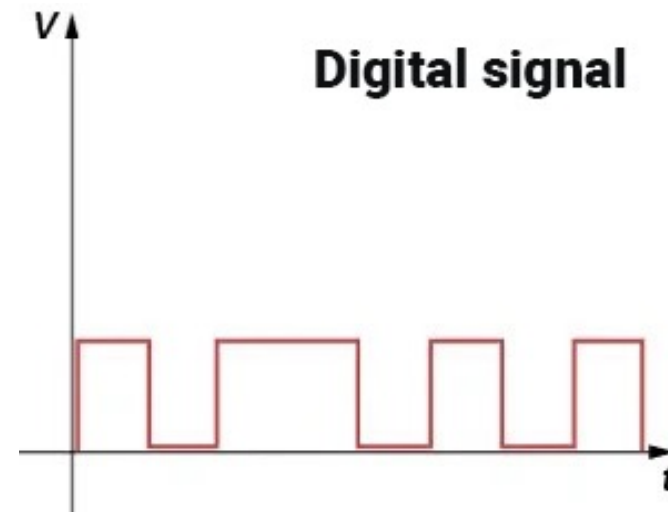
Analog Computing With Neutral Atoms

A Robust Computational Path for the NISQ Era

Analog VS Digital



$[0,1]$



$\{0,1\}$

Analog Computing VS Digital Computing

Classical

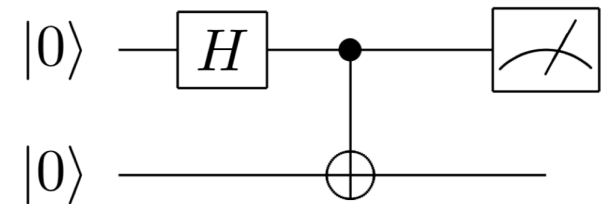
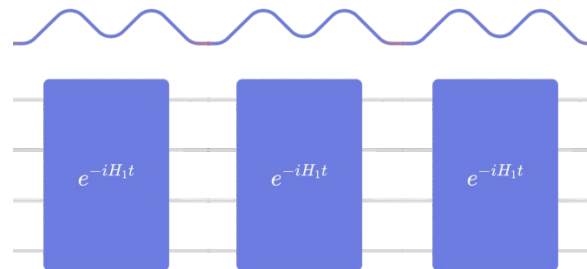
$$s(t)$$

$\{AND, NOT, \dots\}$

Quantum

$$\mathcal{H}(t)$$

$\{H, CNOT, \dots\}$



The Hamiltonian $\mathcal{H}(t)$

- Physically, the **energy** of the system
- A **Hermitian** operator
- An **Observable**
- It governs how the quantum state **evolves**:

$$\mathcal{H} = \mathcal{H}^\dagger$$

$$i\hbar \frac{\partial |\psi\rangle}{\partial t} = \mathcal{H}|\psi\rangle$$

If the Hamiltonian doesn't depend on time the equation becomes:

$$|\psi\rangle(t) = e^{-it\mathcal{H}} |\psi\rangle_0$$

Evolved state Unitary operation Initial state

$$|\Psi\rangle(t) = e^{-it\kappa} |\Psi\rangle_0$$

Digital computing:

We fix κ to obtain quantum gates:

- $e^{-it_1\kappa_1} = H$
- $e^{-it_2\kappa_2} = X$
- $e^{-it_3\kappa_3} = \text{CNOT}$

Analog computing:

We continuously change κ in time depending on the problem:

$$\kappa(t)$$

Why Analog Computing?

Analog Computing

Advantages:

- **High flexibility** (hardware efficiency);
- Noise Resilience;
- Scalability for simulations.

Cons:

- Lack of Universality;
- Hardware-specific knowledge needed;
- No quantum error correction.

Digital Computing

Advantages:

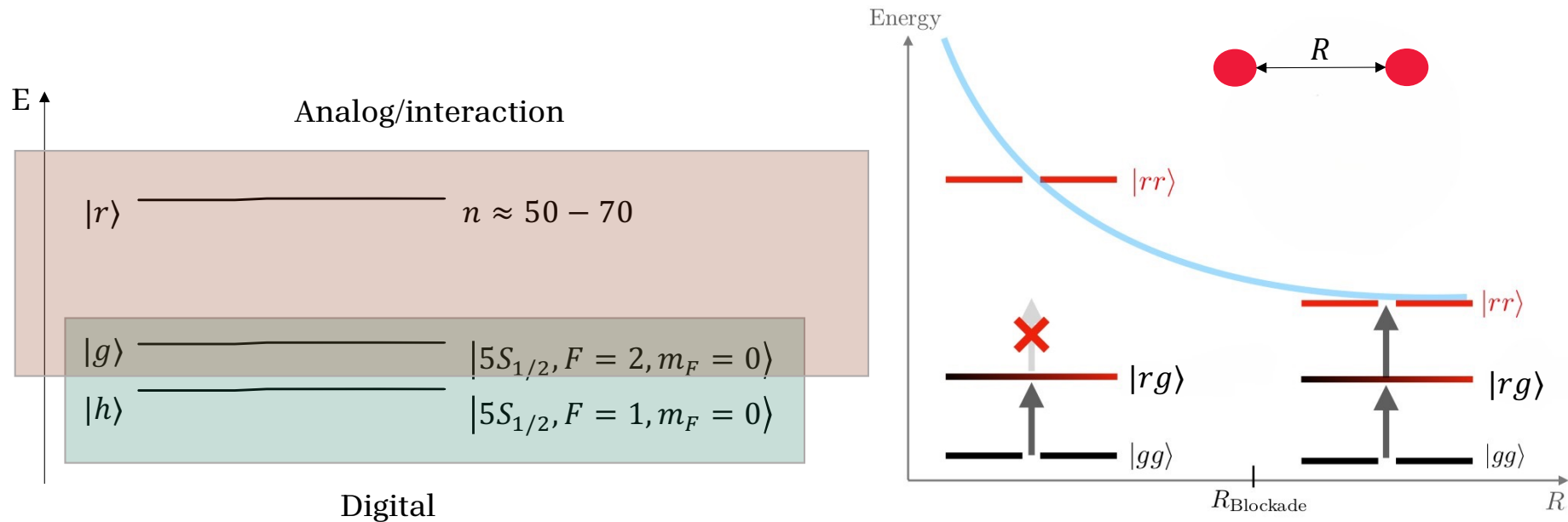
- **Universality**;
- High abstraction level;
- Potential Quantum error correction.

Cons:

- High gate overhead;
- Higher sensitivity to noise;
- Scalability issues for the NISQ era.

Neutral Atoms

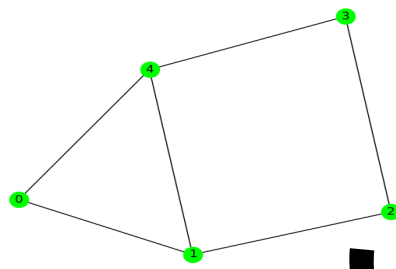
- **Qubit**: encoded in the lower two energy states ($|g\rangle, |h\rangle$)
- **Interaction between qubits**: using Rydberg states (high energy states $|r\rangle$)



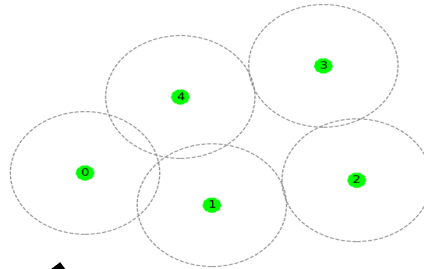
Why Neutral Atoms?

- **Natural encoding** of graphs (node $\rightarrow$ atom)
- Intrinsic **long-range interaction**

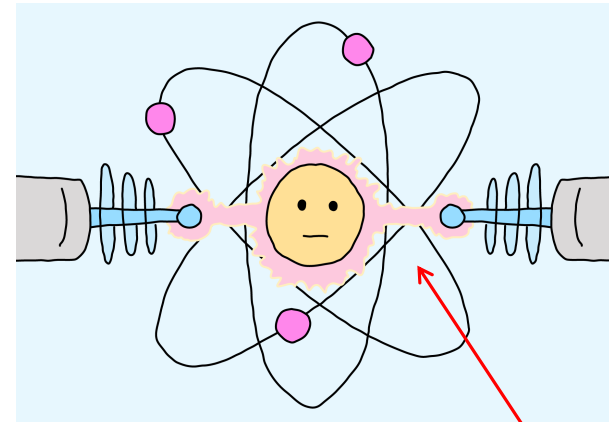
1. Original graph



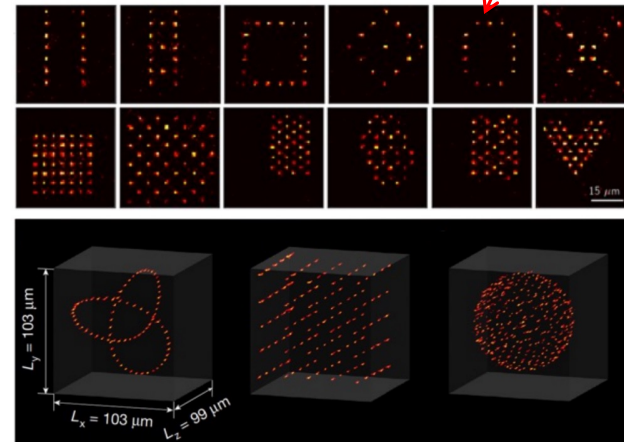
3. Atoms' positions



2. Positioning based on graph edges and device constraints



Optical tweezers trap neutral atoms



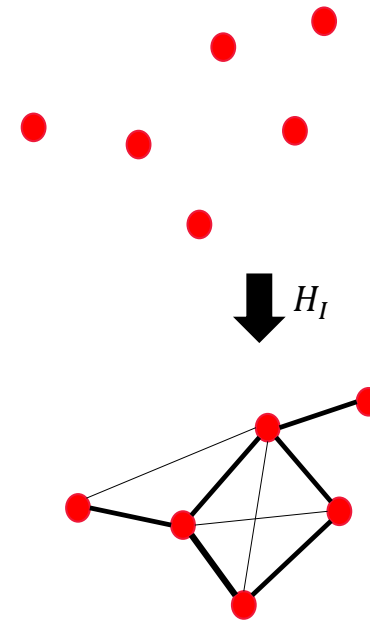
Neutral-Atom Quantum Dynamics Driven by Laser Pulses

Quantum evolution can be described with Hamiltonian formalism:

$$U(t)|\psi_0\rangle = e^{-iHt}|\psi_0\rangle$$

$$H = \sum_{i < j = 1}^N \underbrace{V_{ij}|\mathbf{r}\rangle_i\langle\mathbf{r}| \otimes |\mathbf{r}\rangle_j\langle\mathbf{r}|}_{\text{Interaction term } H_I}$$

- $V_{ij} = \text{Const}/R_{ij}^6$ is the interaction strength, where R_{ij} is the distance between atom i and atom j
- N is the number of atoms in the system

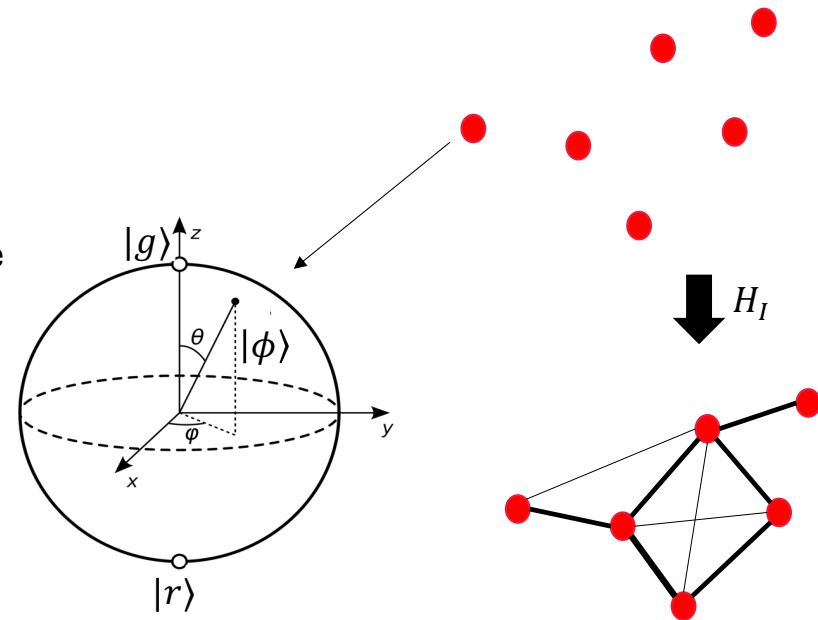


Neutral-Atom Quantum Dynamics Driven by Laser Pulses

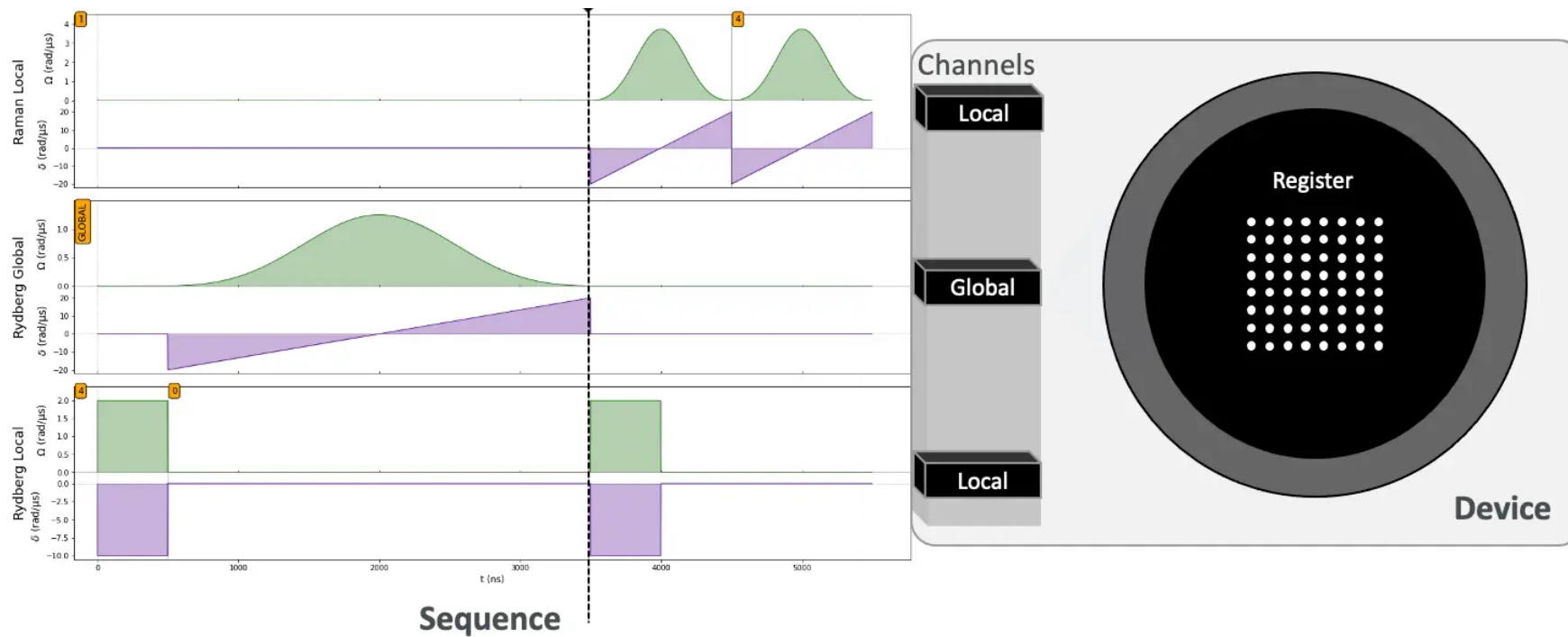
$$H = H_I + \Omega \sum_{k=1}^N \hat{n} \cdot \sigma_{gr}^{(k)}$$

*Global rotations
on the Bloch spheres*

- $\sigma_{gr}^{(k)} = (\sigma_x^{(k)}, \sigma_y^{(k)}, \sigma_z^{(k)})$: Pauli vector, acting on the two-level subspace $\text{span}\{|r\rangle, |g\rangle\}$ of atom k
- Ω : controls the rotation angle
- $\hat{n}$: unit vector defining the rotation axis



Quantum Analog Computing



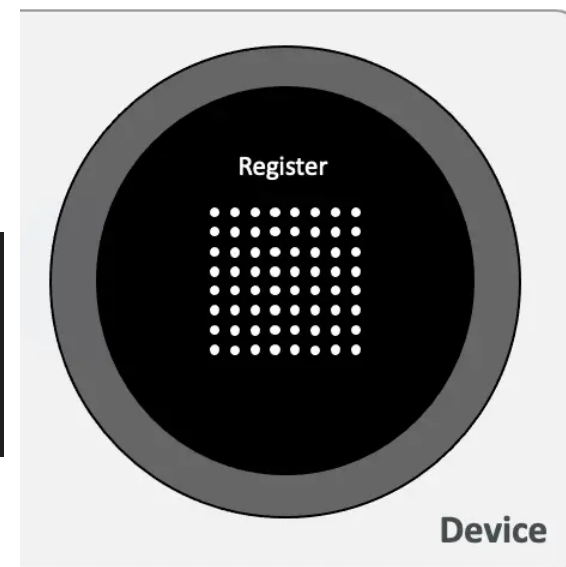
<https://docs.pasqal.com/pulser/>

Quantum Analog Computing

- **The device** imposes **constraints on the topology** that the register can define.
- **The register** defines the **spatial arrangement** and **connectivity** of the qubits in the neutral-atom device.

```
import pulser
print(pulser.AnalogDevice._specs()) # DigitalAnalogDevice

qubits = {"q_1":[0, 0], "q_2":[0, 1], "q_3":[1, 0], "q_4":[1, 1]}
register = pulser.Register(qubits)
register.draw()
```



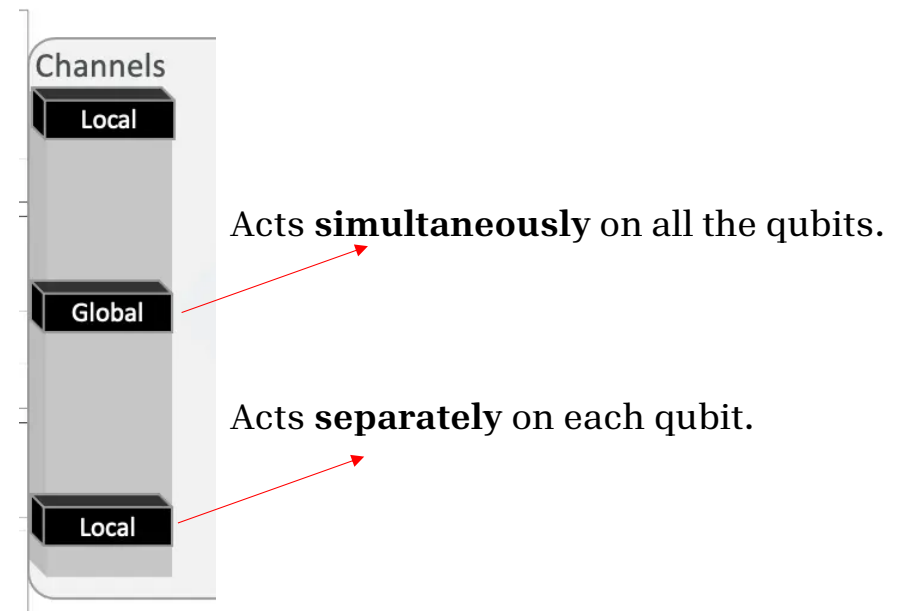
Quantum Analog Computing

- **The Channels** specify the **physical lasers** that we are using on the register. The available channel types depend on the device.

```
sequence = pulser.Sequence(register,  
pulser.AnalogDevice)  
sequence.declare_channel("RG", "rydberg_global")
```

$$H = H_I + \Omega \sum_{k=1}^N \hat{n} \cdot \sigma_{gr}^{(k)}$$

*Global rotations
on the Bloch spheres*



Quantum Analog Computing

- **The sequences** specify the **evolution of the parameters** in time:

- Rabi frequency Ω ; 
- Detuning δ ; 
- Phase ϕ .

```
import numpy as np
time = 1000 # Time in ns

Omega = np.pi # Rabi frequency in MHz

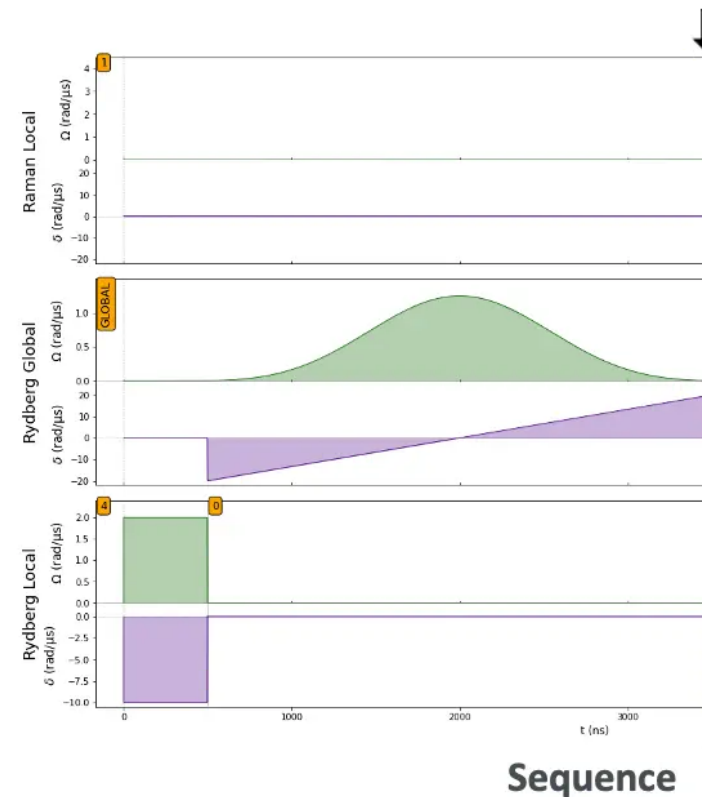
detuning = 0

phase = 0

pulse = pulser.Pulse.ConstantPulse(time, Omega, detuning, phase)
sequence.add(pulse, "RG") # Add the pulse on the RG channel

sequence.draw()
```

<https://docs.pasqal.com/pulser/>



QUBO problems

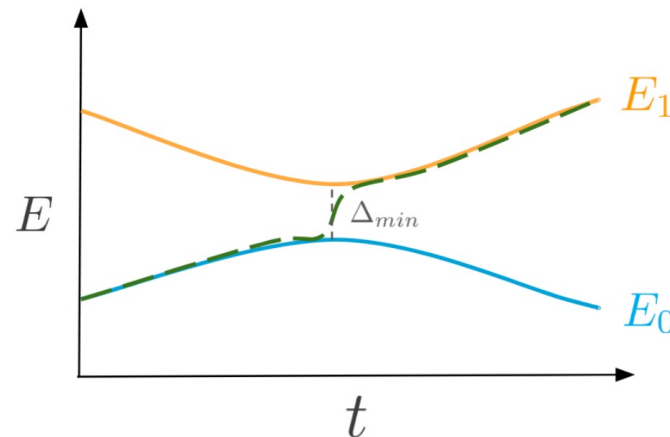
$$f(z) = z^T Q z \quad \text{where } Q \in \mathbb{R}^n \times \mathbb{R}^n \text{ and } z \in (\mathbb{Z}_2)^n$$

- We want $z^* = \arg \min_{z \in \mathcal{Z}} f(z)$
- QUBO is NP-hard $\longrightarrow$ We need **heuristic methods** (simulated annealing...)
- All NP problems can be mapped to QUBO (in polynomial time)

Quantum **Adiabatic** Algorithm (QAA)

Quantum adiabatic theorem:

*A **physical system** remains in its **instantaneous eigenstate** if a given perturbation is acting on it **slowly enough** and if there is a gap between the eigenvalue and the rest of the Hamiltonian's spectrum.*

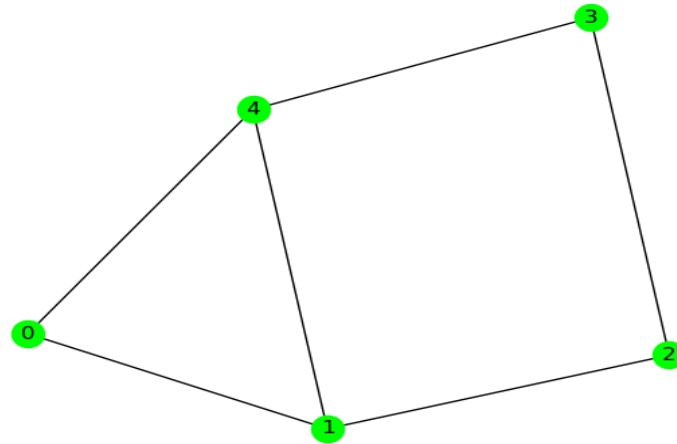


Mapping the Matrix to the Hamiltonian

$$Q \xrightarrow{\text{red arrow}} \frac{H_Q}{\hbar}(t) = \sum_{k=1}^N \left(\frac{\Omega(t)}{2} e^{-i\phi(t)} |g\rangle\langle r|_k + \frac{\Omega(t)}{2} e^{i\phi(t)} |r\rangle\langle g|_k \underbrace{- \delta(t) |r\rangle\langle r|_k}_{\text{Off-diagonal entries of } Q} + \underbrace{\sum_{j < k} Q_{kj} \hat{n}_k \hat{n}_j}_{\text{Off-diagonal entries of } Q} \right)$$

Diagonal entries of Q

Off-diagonal entries of Q



Quantum **Adiabatic** Algorithm (QAA)

- Initialize the system in the **ground state** of the starting Hamiltonian

$$H(t=0)=H_I \rightarrow |ggggg\rangle = |00000\rangle$$

- Prepare a superposition state by increasing the term Ω with low detuning ($\delta \ll 0$) to detuning $\delta \approx 0$

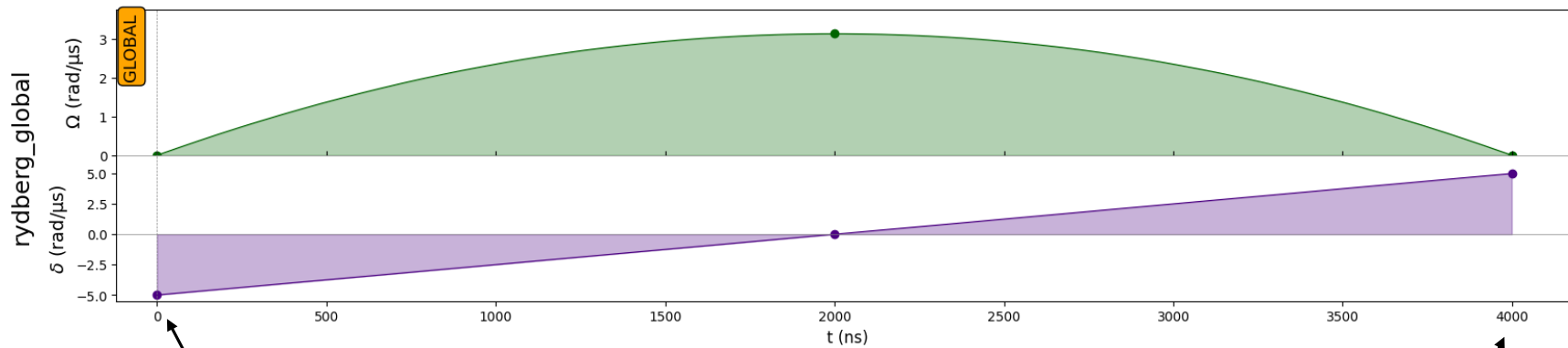
$$H(t=t_m)=H_I+H(\delta, \Omega) \rightarrow |+++++\rangle$$

- Lowering the term Ω to 0 and adjust the detuning to the target values based on Q.

$$H(t=t_f)=H_{QUBO} \rightarrow |\psi(t_f)\rangle = \sum c_x |x\rangle$$

Where the probabilities ($|c_x|^2$) of the optimal strings are dominant.

Quantum **Adiabatic** Algorithm (QAA)



$$H(t=0)=H_I \rightarrow |ggggg\rangle = |00000\rangle$$

$$H(t=t_m)=H_I+H(\delta,\Omega) \rightarrow |+++++\rangle$$

$$H(t=t_f)=H_{\text{QUBO}} \rightarrow |\psi(t_f)\rangle = \sum c_x |x\rangle$$

Q&A